SCIENCE QUESTION PAPER

PHYSICS – CHAPTER: SOUND

Class: IX
Time Allowed: 45 Minutes
Maximum Marks: 25

General Instructions:

1. All questions are compulsory.
2. Marks for each question are indicated against it.
3. Draw neat and well-labeled diagrams wherever necessary.

Q1. Sound Propagation and Wave Properties (5 Marks)

Explain how sound is produced by a vibrating source and propagates through a medium to reach an observer. Define the following physical quantities associated with a sound wave:

(a) Amplitude
(b) Frequency
(c) Wavelength
(d) Time Period

Answer Hint Box (Strategy for Full Marks):

- **Mechanism:** Your explanation must include the terms **Compressions** (regions of high pressure) and **Rarefactions** (regions of low pressure).
- **Medium:** Clearly state that sound is a *mechanical wave* and requires a material medium (solid, liquid, or gas) for propagation; it cannot travel in a vacuum.
- **Definitions:** Provide precise definitions for each term and mention their **SI units**:
- Amplitude – meter (m)
- Frequency – hertz (Hz)
- Wavelength – meter (m)
- Time period – second (s)
- **Diagram:** For full marks, draw a simple wave diagram and label crest, trough, amplitude, and wavelength.

Q2. The Physics of Echoes (4 Marks)

(a) What is an echo?
(b) Calculate the minimum distance required between a sound source and a reflector to hear a distinct echo when the temperature is **20°C**. (Speed of sound in air = **344 m/s**).
(c) Explain how this minimum distance would be affected if the temperature of the air were to rise.

Answer Hint Box (Strategy for Full Marks):

- **Core Concept:** Begin by stating the principle of **persistence of hearing**, i.e., the human ear retains sound for approximately **0.1 s**.
- **Formula & Calculation:**
- Use the relation: $2d = v \times t$
- $d = (344 \times 0.1)/2 = 17.2\,m$
- **Temperature Effect:** As temperature increases, the speed of sound increases. Hence, the **minimum distance required to hear an echo also increases**.

Q3. Acoustic Engineering and Reverberation (5 Marks)

Define reverberation. Explain why excessive reverberation is considered a defect in the design of large auditoriums. Describe **two methods** used to control it. Also explain the purpose of **curved ceilings** in concert halls.

Answer Hint Box (Strategy for Full Marks):

- **Definition:** Reverberation is the persistence of sound in an enclosed space due to multiple reflections from surfaces like walls, ceilings, and floors.
- **Problem:** Excessive reverberation causes overlapping sounds, making speech and music unclear.
- **Solutions:** Use sound-absorbing materials such as:
- Compressed fibreboard
- Rough plaster
- Curtains, carpets, or acoustic panels
- **Curved Ceilings:** They help in uniform distribution of sound waves, prevent dead spots, and improve overall sound quality.

Q4. SONAR and Ultrasound Applications (6 Marks)

(a) What is the full form of SONAR?
(b) With the help of a neat, labeled diagram, explain the working principle of SONAR to determine the depth of the sea.
(c) State **one industrial application** of ultrasound.

Answer Hint Box (Strategy for Full Marks):

- **Acronym:** SONAR stands for **Sound Navigation And Ranging**.
- **Working Principle:**
- SONAR uses a **transmitter** and a **receiver**.
- The transmitter emits ultrasonic waves which travel through water, reflect from the seabed, and return to the receiver.
- The time taken by the waves is used to calculate the depth of the sea.
- **Diagram:** Draw a ship on the water surface with downward and upward arrows showing transmitted and reflected waves.
- **Industrial Application:**
- Detecting cracks and flaws in metal blocks
- Ultrasonic cleaning of complex machinery parts

Q5. Critical Thinking and Numerical Analysis (5 Marks)

Two children are standing at opposite ends of a **1 km long aluminum rod**. One child strikes one end of the rod with a stone. The second child hears **two distinct sounds**.

(a) Explain why two sounds are heard.
(b) Calculate the **ratio of time taken** by sound to travel through air and through aluminum.

(Given: Speed of sound in air = **346 m/s**; Speed of sound in aluminum = **6420 m/s**)

Answer Hint Box (Strategy for Full Marks):

- **Reasoning:** Sound travels at different speeds in different media. Sound travels much faster through solid aluminum than through air, so the sound through the rod reaches first.
- **Calculation:**
- $t_{air} = 1000/346$
- $t_{aluminum} = 1000/6420$
- Ratio: $t_{air}/t_{aluminum} = 6420/346 \approx 18.55$
- **Final Answer:** The ratio of time taken through air to aluminum is **approximately 18.55 : 1**.

End of Question Paper